

RentIt

"Why Buy When You Can Rent?"

A Secure Peer-to-Peer Rental Marketplace
Major Project Presentation

The Problem: Idle Assets

UNDERUTILIZED EQUIPMENT

People purchase expensive items (DSLRs, power tools, tents) for one-time use. These valuable assets sit idle in homes for the majority of their lifespan.

HIGH COSTS FOR RENTERS

Others need these items temporarily but cannot afford to buy them new. Existing rental shops are expensive, limited in stock, and lack peer-to-peer trust.

Our Solution: RentIt

A full-stack MERN platform facilitating secure peer-to-peer rentals.

FOR OWNERS

List idle assets seamlessly. Review requests, approve renters, and generate passive income safely.

FOR RENTERS

Search categories, request bookings, and access necessary items temporarily at a fraction of retail cost.

Target Audience

- **Students:** Need cameras, projectors, or event gear cheaply.
- **Freelancers:** Photographers needing specialized lenses.
- **Travelers:** Needing camping tents for a weekend.
- **Homeowners:** Occasional DIYers needing power tools.

A hyper-local marketplace catering to short-term community needs.

Architecture & Stack

FRONTEND UI

React.js (Vercel Edge Network)

BACKEND API

Node.js & Express (Vercel Serverless)

DATABASE

MongoDB Atlas (Cloud Cluster)

STORAGE

Cloudinary (Image Hosting)

SECURITY

JWT & Bcrypt Hashing

STYLING

Custom CSS (Glassmorphism)

Database Schema

Relational references within a NoSQL document structure.

USERS

Stores names, emails, encrypted passwords, and user IDs.

ITEMS

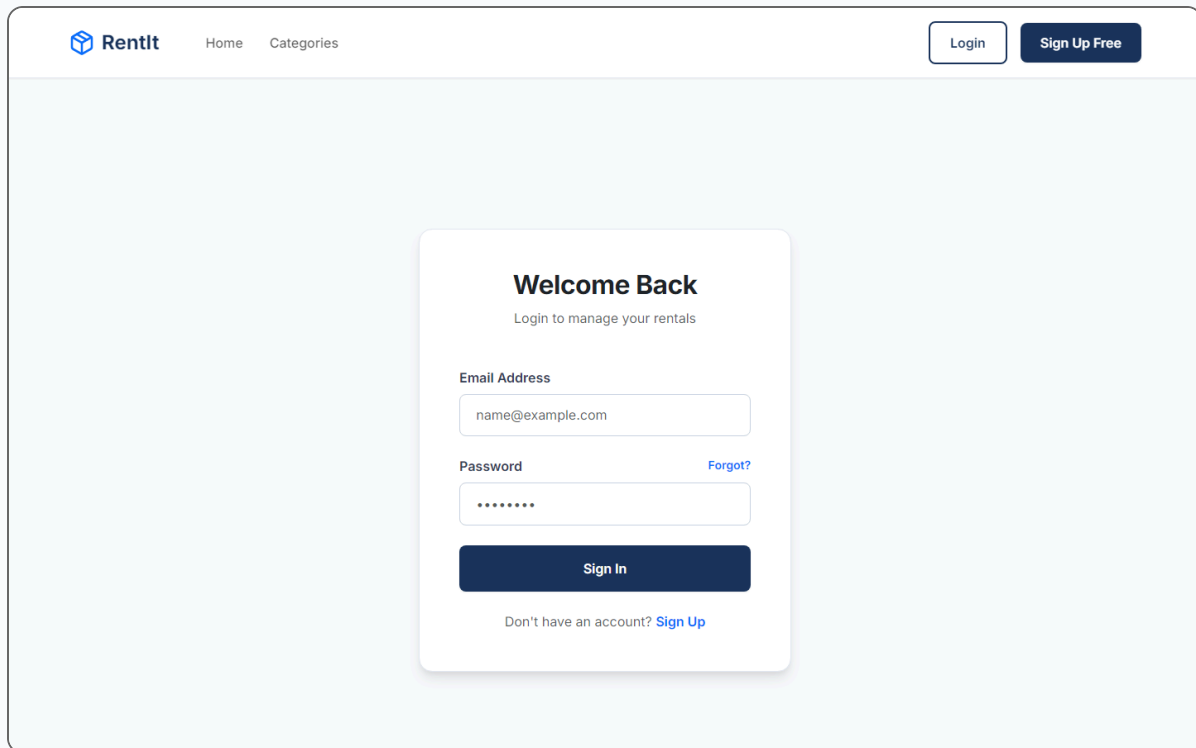
Asset title, price, description, images, and Owner ID ref.

BOOKINGS

Start/End dates, status (Pending/Approved), Renter & Item references.

1. Secure Login

Users (both renters and owners) authenticate using encrypted credentials. A JWT token is stored securely to maintain the session.



The image shows a web application interface for 'RentIt'. The header includes the 'RentIt' logo, navigation links for 'Home' and 'Categories', and buttons for 'Login' and 'Sign Up Free'. The main content area features a 'Welcome Back' login form. The form has a title 'Welcome Back' and a subtitle 'Login to manage your rentals'. It contains two input fields: 'Email Address' with the placeholder 'name@example.com' and 'Password' with masked characters '*****'. A 'Forgot?' link is positioned to the right of the password field. A dark blue 'Sign In' button is located below the password field. At the bottom of the form, there is a link that says 'Don't have an account? Sign Up'.

RentIt Home Categories Login Sign Up Free

Welcome Back

Login to manage your rentals

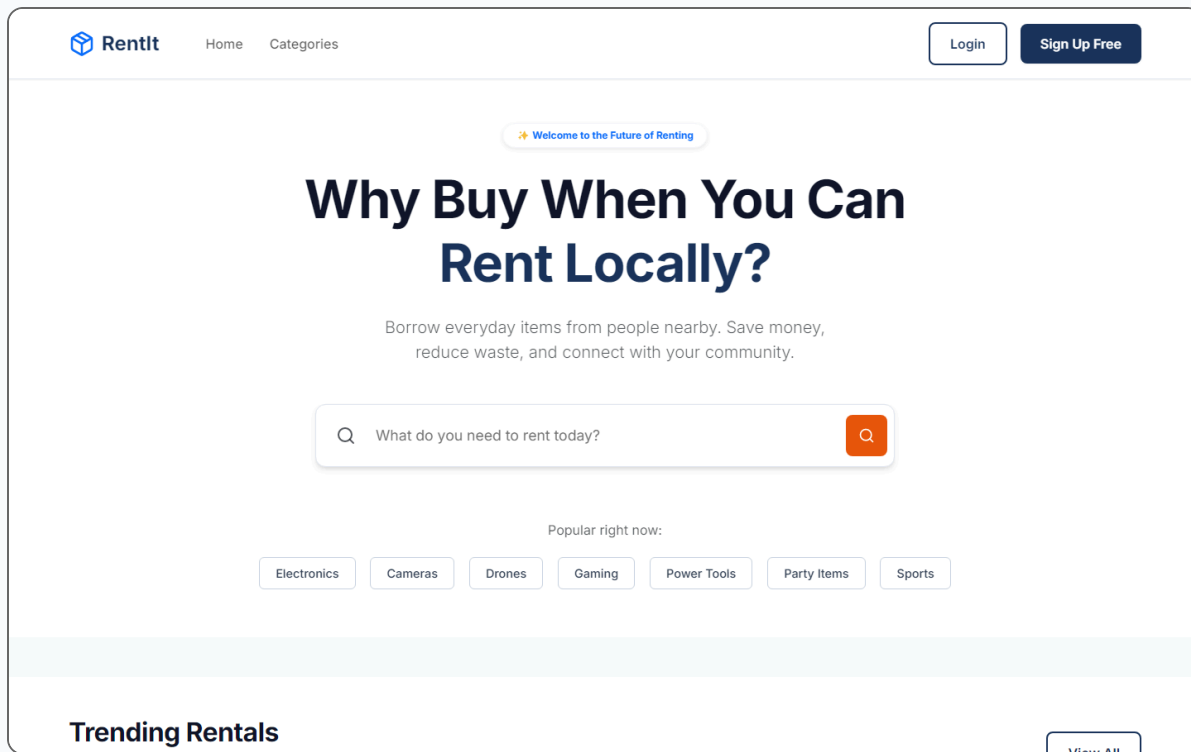
Email Address
name@example.com

Password [Forgot?](#)

Sign In

Don't have an account? [Sign Up](#)

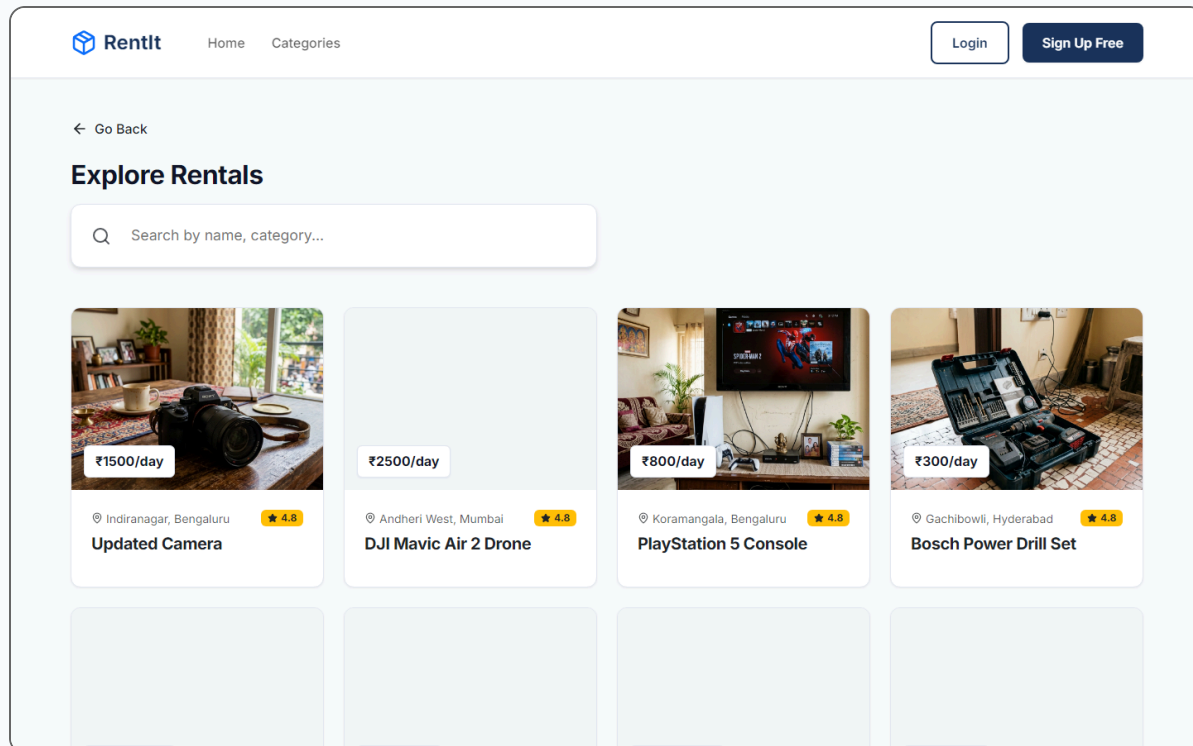
2. Item Discovery



The React frontend displays available items dynamically. Users can browse pricing, locations, and high-quality images.

3. Search & Filters

Renters can narrow down inventory quickly using search terms (e.g. "tents") or category filters, syncing with URL parameters.



4. Item Details

[Home](#) [Dashboard](#) [Categories](#)

[Edit Profile](#) [Logout](#)

[← Go Back](#)



Updated Camera

📍 Indiranagar, Bengaluru ⭐ 4.8 (12 reviews)

Description

₹1500 / day

Pickup Date

Return Date

[Request to Book](#)

You won't be charged yet

Clicking on an item reveals in-depth information, descriptions, user reviews, and the form required to request a booking.

5. Creating a Request

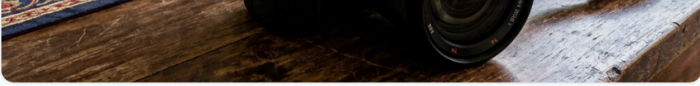
Renters easily select their desired rental dates on the calendar and submit a booking request securely to the item's owner.

 RentIt

[Home](#) [Dashboard](#) [Categories](#)

Edit Profile

Logout



Updated Camera

Indiranagar, Bengaluru

★ 4.8 (12 reviews)

Description

desc

OU

Owned by Owner User

Identity Verified

Pick up location: Indiranagar, Bengaluru

Reviews from Renters

Renter User

"Absolutely amazing! The camera was in perfect condition and the owner was very helpful."

★ 5/5

₹1500 / day

Pickup Date

dd-----yyyy

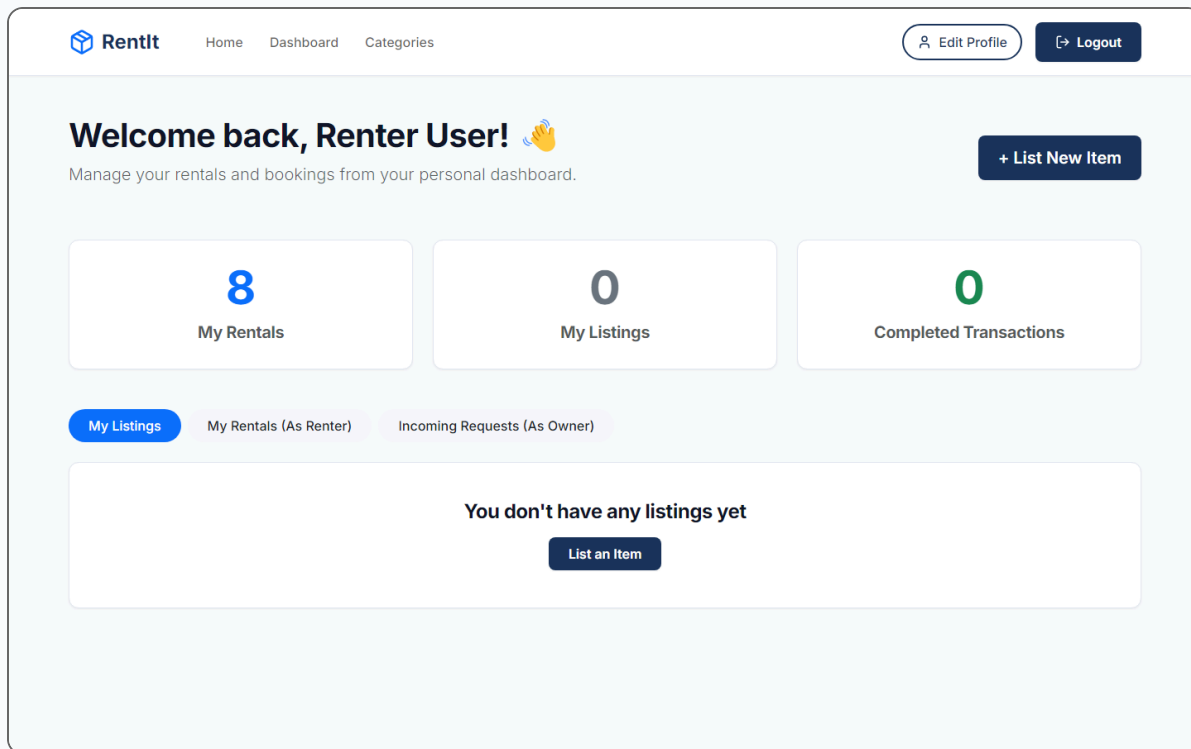
Return Date

dd-----yyyy

Request to Book

You won't be charged yet

6. Renter Dashboard



Renters can track all their active and pending rental requests, cancel bookings, and leave reviews upon completion.

7. Creating a Listing

Owners utilize a clean, responsive form to add new assets. Data is validated on the client side, and images are pushed directly to Cloudinary.

 **RentIt**

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[← Go Back](#)

List a New Item

Title

Description

Category

Electronics

Pick Up Location (City, Area)

e.g. Indiranagar, Bengaluru

Optional Google Maps Link

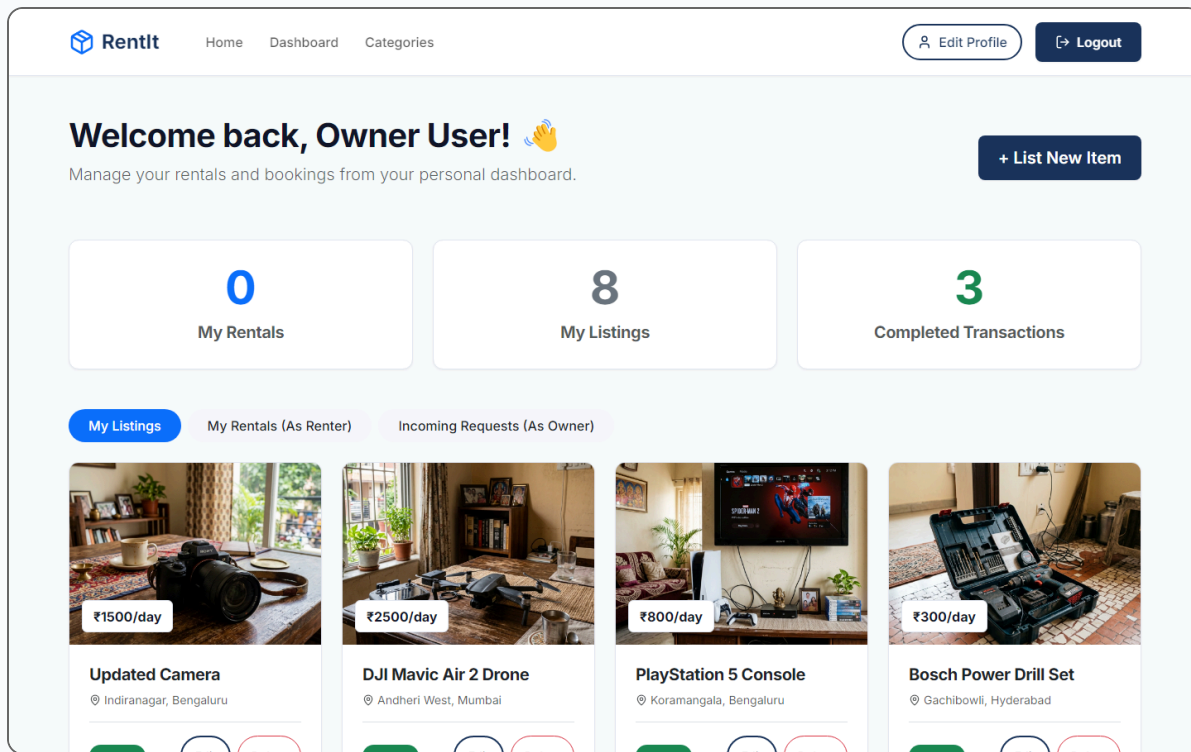
Contact Number (For Renters)

e.g. 9876543210

Price per Day (₹)

Security Deposit (₹)

8. Owner Dashboard



The centralized hub. Owners can track active listings and instantly approve or reject incoming booking requests from renters.

Thank You

Empowering local communities through sustainable sharing.

Live Platform:

rentit-marketplace.vercel.app

Source Code:

github.com/mscharan1810/RentIt